Warehouse Case Study

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Executive Summary

Super Shopper is facing two main inventory management issues that this report will explore and address. First, high-demand products are not refilled fast enough and often out of stock. Second, low-demand products are overstocked and end up either expiring or being returned. The goal of this analysis is to identify inefficiencies in the inventory management process and suggest alternative procedures that could decrease the likelihood of these issues reoccurring. Analysis techniques will be applied to Store 1 so that the process can be repeated for Stores 2-10.

Key Insights

- **Stockouts:** The following products experienced zero inventory over several weeks, and are prime candidates for an increase in reorder amounts or a re-evaluation of current reorder points.

Product	Notes	Total Weeks Stocked Out
Dishwasher Soap	Out of stock weeks 2, 3 & 6-7	4
Scouring Pads	Out of stock weeks 3 & 5-8	5
Window Cleaner	Out of stock weeks 1, 3 & 8	3
Hand Soap	Out of stock weeks 2, 4-5 & 7	4
Toilet Paper	Out of stock weeks 1, 4 & 7	3
Vitamins	Out of stock weeks 1, 3 & 7	3
Antacid	Out of stock weeks 3-4 & 8	3
Motion Sickness	Out of stock weeks 2-3 & 7	3
Ink Refills	Out of stock weeks 2-5	4
Notebooks	Out of stock weeks 2-4 & 8	4

Staplers	Out of stock weeks 1, 3-4 & 7	4

- **Overstocks:** These products experience very little turnover due to low customer interest, inaccurate demand forecasts or misaligned reorder points, and could benefit from better alignment with the warehouse and enhanced marketing efforts.

Product	Notes
Toothpaste	Never drops below 10. Starts at 20, finishes at 10
Cleaning Solvent	Completely flat. Stays at 15 for all 8 weeks
Pencils	Completely flat. Stays at 20 for all 8 weeks

- **Warehouse Inventory & Reorder Points:** Comparing inventory management at the store level to the warehouse level for products experiencing inefficiencies. Products experiencing frequent stockouts should have their warehouse inventory levels and reorder points increased, while products experiencing overstock at the store level should have their warehouse level inventory and reorder point decreased. Products experiencing stockouts over consecutive weeks further suggest procedural delays at the warehouse level which can be offset with increased thresholds.

Product	Issue	Warehouse Inventory	Reorder Point	Notes
Dishwasher Soap	Frequent Stockout	50	25	Increase inventory level and reorder point

Scouring Pads	Frequent Stockout	100	50	Increase inventory level and reorder point
Window Cleaner	Frequent Stockout	100	50	Increase inventory level and reorder point
Hand Soap	Frequent Stockout	100	50	Increase inventory level and reorder point
Toilet Paper	Frequent Stockout	100	50	Increase inventory level and reorder point
Vitamins	Frequent Stockout	100	50	Increase inventory level and reorder point
Antacid	Frequent Stockout	40	20	Increase inventory level and reorder point
Motion Sickness	Frequent Stockout	50	25	Increase inventory level and reorder point

Ink Refills	Frequent Stockout	150	100	Increase inventory level and reorder point
Notebooks	Frequent Stockout	50	25	Increase inventory level and reorder point
Staplers	Frequent Stockout	20	10	Increase inventory level and reorder point
Toothpaste	Overstocked	100	50	Reduce inventory levels and reorder points
Cleaning Solvent	Overstocked	50	25	Reduce inventory levels and reorder points
Pencils	Overstocked	100	50	Reduce inventory levels and reorder points

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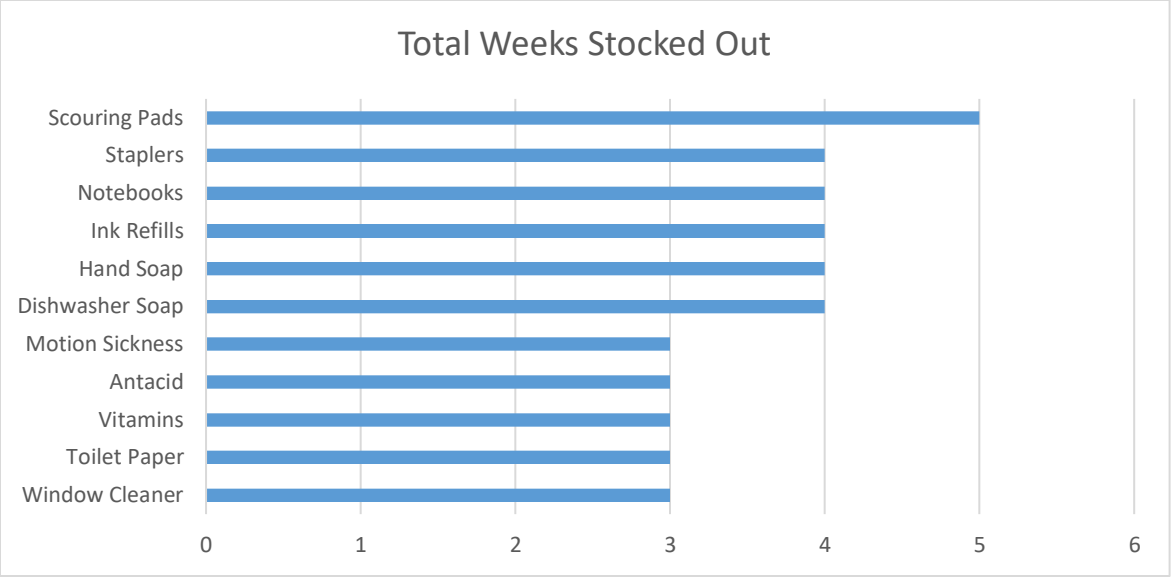
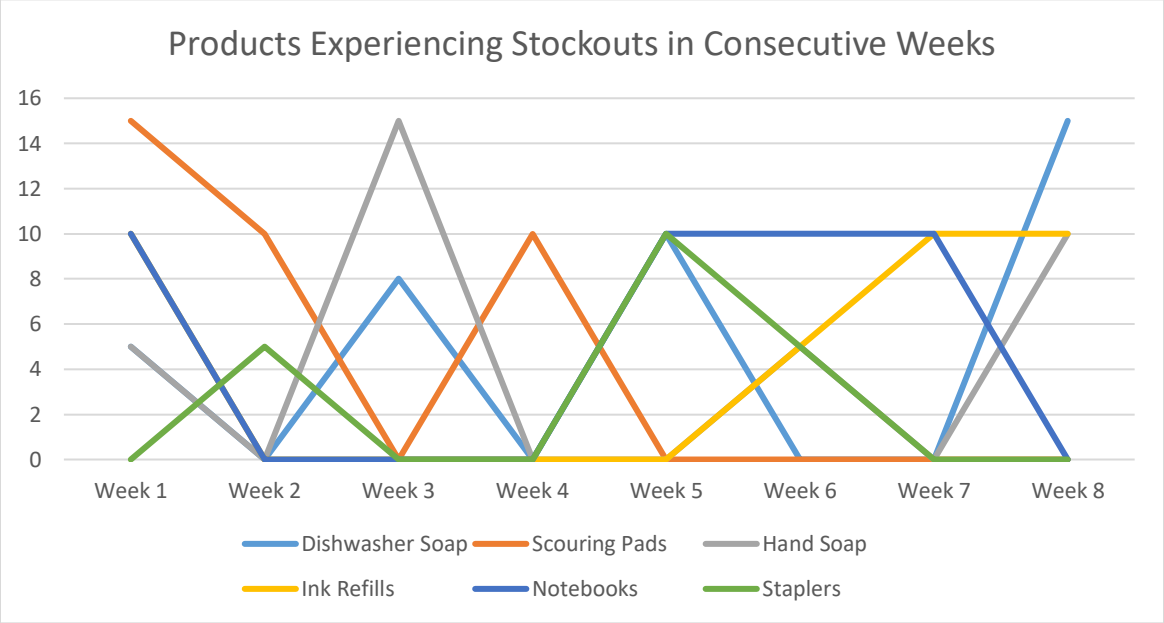
Visualizations & Recommendations



11 products experienced stockouts for at least 3 weeks or more. Considering the length of the sample (8-weeks), this means 44% of the store's products were unavailable to customers nearly 38% of the time. This suggests a holistic misalignment with the current forecasted inventory demand as stock out issues are widespread. Static warehouse inventory handling procedures should be switched to a dynamic, store-specific model to better match store-level demand.



Of the 11 products that experienced frequent stockouts, 6 products experienced stockouts during at least 2 consecutive weeks. This suggests a delay in warehouse ordering and shipping procedures that must be accounted for, further enhancing the decision to increase reorder points. If reorder points are increased, products will be ordered while inventory levels remain high, offsetting any delays in shipping and receiving.

Product	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Dish Soap	15	12	10	15	0	0	10	5
Dishwasher Soap	5	0	8	0	10	0	0	15
Scouring Pads	15	10	0	10	0	0	0	0
Window Cleaner	0	10	0	10	5	10	10	0
Paper Towels	15	0	5	10	10	0	15	20
Trash Bags	10	15	10	5	0	0	15	15
Toothpaste	10	15	15	15	15	10	15	10
Mouthwash	15	10	5	0	10	10	10	10
Hand Soap	5	0	15	0	0	5	0	10
Shampoo	12	10	5	0	10	5	10	10
Hand Lotion	5	15	12	0	0	10	5	15
Toilet Paper	0	15	10	0	15	20	0	15
Cleaning Solvent	15	15	15	15	15	15	15	15
Asprin	15	15	15	15	5	12	15	15
Vitamins	0	15	0	10	15	15	0	10
Antacid	10	10	0	0	5	5	10	0
Laxatives	10	10	10	5	0	10	0	10
Motion Sickness	5	0	0	10	10	5	0	5
Pens	15	15	15	5	15	15	12	0
Ink Refills	10	0	0	0	0	5	10	10
Pencils	20	20	20	20	20	20	20	20
Notebooks	10	0	0	0	10	10	10	0
Paper	12	0	15	10	5	0	15	10
Staplers	0	5	0	0	10	5	0	0
Paper Clips	10	5	0	10	5	5	5	10

In the above heat map, darker shades of red represent dangerously low levels of inventory, while darker shades of blue represent dangerously high levels of inventory. Optimal inventory levels are represented in white, or with no shading. Looking at the total inventory, very few products maintain optimal levels. Most of the inventory is either overstocked or

understocked. This visualization clearly communicates the need for developing and implementing a dynamic forecasting system.

According to the Accountant's section of the Warehouse Reorder Point dataset, every single product operates at a 20% profit margin. For profit margins this thin, there is not much room to allow for waste acceptance. It is always crucial to optimize inventory management, but unless pricing structures are re-developed, the margins that Super Shopper operates within increases the impact that mismanagement can have on the organization.